KERALA DIPLOMA CURRICULUM REVISION 2026

Course 3091 | 3 Credits: 3 | Program Core | Source-grounded

Natural Rubber Production

 To provide an idea about different rubber yielding plants, their propagation methods

OFFICIAL SOURCE DECLARATION

How this lesson was prepared

The supplied official SITTTR syllabus PDF for Course 3091 is the authoritative source for course information, objectives, course outcomes, CO–PO mapping, modules, topics, activities, assessment structure and references. Explanations, study prompts and Malayalam support are educational elaboration and are not presented as additional official syllabus text.

Source file: 3091.pdf from the uploaded official SITTTR syllabus archive. **Course code and title:** preserved from the source.

COURSE DASHBOARD

Official course information

Course information

Item	Official information
Programme	Diploma in Polymer Technology
Course code	3091
Course title	Natural Rubber Production
Semester	3 Credits: 3
Course category	Program Core
Periods per week	3 (L:2, T:1, P:0) Periods per semester: 45

Item	Official information
Periods per semester	45
Credits	3
Prerequisite	See the official syllabus PDF
Assessment	Use the official assessment section reproduced below

Modules

4 official module sections detected.

Topic entries

17 source-aligned topic entries retained.

Study mode

Theory and application emphasis.

STUDY METHOD

How to use this lesson

First pass

Read the official source declaration, dashboard and module transcript. Mark unfamiliar terms without changing the official terminology.

Second pass

Use each topic card to build a definition, principle, steps or components, application and exam answer. Attempt the Q&A before opening the answers.

Practical evidence

Where the course is laboratory or workshop based, maintain an observation notebook with procedure, instruments, readings, safety points and conclusion.

Revision pass

Return to the source excerpt, coverage table, activities and model paper. The generated paper is practice material, not an official examination paper.

LEARNING OUTCOMES

Objectives, outcomes and mapping

Course objectives

- To provide an idea about different rubber yielding plants, their propagation methods
- and exposure to various latex extraction methods.
- To give an idea about preservation and concentration of NR latex.
- To get an idea about the dry marketable forms and modified forms of Natural Rubber.

Course outcomes

CO	Official outcome	Study level
CO1	11 Understanding propagation methods and extraction of NR Explain the composition, preservation and	See official syllabus
CO2	12 Understanding concentration of NR latex Explain the processing of NR latex in to dry	See official syllabus
CO3	10 Understanding marketable forms Describe the production, properties and	See official syllabus
CO4	applications of specialty rubbers and reclaimed 10 Understanding rubber	See official syllabus

CO–PO mapping evidence

Row	Extracted official mapping
CO1	CO1 3
CO2	CO2 3
CO3	CO3 3
CO4	CO4 3

COVERAGE AUDIT

Complete syllabus coverage

Every detected official module is represented below. Each module contains topic cards and an expandable official syllabus transcript to preserve source terminology.

Module coverage

Module	Official heading	Topic entries	Hours
Module 1	extraction of NR	4	As specified
Module 2	Explain the composition, preservation and concentration of NR latex	4	As specified
Module 3	Explain the processing of NR latex in to dry marketable forms	4	As specified
Module 4	and reclaimed rubber	5	As specified

MODULE 1

extraction of NR

This module is presented from the official syllabus scope for Natural Rubber Production. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 1 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: extraction of NR
- Official duration: As specified

- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

Explain the importance and uniqueness of NR 2 Remembering Describe various propagation methods and

Explain the importance and uniqueness of NR 2 Remembering Describe various propagation methods and is an official topic in Module 1 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Explain the importance and uniqueness of NR 2 Remembering Describe various propagation methods and using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, explain the importance and uniqueness of nr 2 remembering describe various propagation methods and should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Explain the importance and uniqueness of NR 2 Remembering Describe various propagation methods and means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ Explain the importance and uniqueness of NR 2 Remembering Describe various propagation methods and □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

classification of clones in terms of rubber Understanding 3 plants Explain tapping and various processes associated

classification of clones in terms of rubber Understanding 3 plants Explain tapping and various processes associated is an official topic in Module 1 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify classification of clones in terms of rubber Understanding 3 plants Explain tapping and various processes associated using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, classification of clones in terms of rubber understanding 3 plants explain tapping and various processes associated should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What classification of clones in terms of rubber Understanding 3 plants Explain tapping and various processes associated means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-[illegible] classification of clones in terms of rubber Understanding 3 plants Explain tapping and various processes associated [illegible] [illegible] definition/meaning [illegible]. [illegible] [illegible] [illegible], steps, application [illegible] [illegible] [illegible]. Technical terms English-[illegible] [illegible] [illegible].

3 Understanding with extraction of latex from rubber trees. Explain method of application and importance

3 Understanding with extraction of latex from rubber trees. Explain method of application and importance is an official topic in Module 1 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding with extraction of latex from rubber trees. Explain method of application and importance using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding with extraction of latex from rubber trees. explain method of application and importance should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding with extraction of latex from rubber trees. Explain method of application and importance means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding with extraction of latex from rubber trees. Explain method of application and importance □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding of rain guarding and yield stimulants. Natural rubber - uniqueness, structure and properties, major sources statistics of production and consumption - India and World. Rubber yielding plants - propagation methods - brown and green budding - different clones, tapping - standard of tappability - tapping task - tapping rest - tapping system. Types of tapping - Ladder, slaughter and puncture tapping. Tapping systems in India, yield stimulants and rain guarding

3 Understanding of rain guarding and yield stimulants. Natural rubber - uniqueness, structure and properties, major sources statistics of production and consumption - India and World. Rubber yielding plants - propagation methods - brown and green budding - different clones, tapping - standard of tappability - tapping task - tapping rest - tapping system. Types of tapping - Ladder, slaughter and puncture tapping. Tapping systems in India, yield stimulants and rain guarding is an official topic in Module 1 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding of rain guarding and yield stimulants. Natural rubber - uniqueness, structure and properties, major sources statistics of production and consumption - India and World. Rubber yielding plants - propagation methods - brown and green budding - different clones, tapping - standard of tappability - tapping task - tapping rest - tapping system. Types of tapping - Ladder, slaughter and puncture tapping. Tapping systems in India, yield stimulants and rain guarding using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding of rain guarding and yield stimulants. natural rubber - uniqueness, structure and properties, major sources statistics of production and consumption - india and world. rubber yielding plants - propagation methods - brown and green budding - different clones, tapping - standard of tappability - tapping task - tapping rest - tapping system. types of tapping - ladder, slaughter and puncture tapping. tapping systems in india, yield stimulants and rain guarding should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding of rain guarding and yield stimulants. Natural rubber - uniqueness, structure and properties, major sources statistics of production and consumption - India and World. Rubber yielding plants - propagation methods - brown and green budding - different clones, tapping - standard of tappability - tapping task - tapping rest - tapping system. Types of tapping - Ladder, slaughter and puncture tapping. Tapping systems in India, yield stimulants and rain guarding means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 1-□□ 3 Understanding of rain guarding and yield stimulants. Natural rubber - uniqueness, structure and properties, major sources statistics of production and consumption - India and World. Rubber yielding plants - propagation methods - brown and green budding - different clones, tapping - standard of tappability - tapping task - tapping rest - tapping system. Types of tapping - Ladder, slaughter and puncture tapping. Tapping systems in India, yield stimulants and rain guarding □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
extraction of NR
  M 1.01    Explain the importance and uniqueness of NR             2
Remembering
            Describe various propagation methods and
  M 1.02    classification of clones in terms of rubber
Understanding
                                                                    3
            plants
            Explain tapping and various processes associated
  M1.03                                                             3
Understanding
            with extraction of latex from rubber trees.
            Explain method of application and importance
  M1.04                                                             3
Understanding
            of rain guarding and yield stimulants.

Contents:
Natural rubber - uniqueness, structure and properties, major sources statistics
of production
and consumption - India and World. Rubber yielding plants - propagation methods -
brown
and green budding - different clones, tapping - standard of tappability - tapping
task -
tapping rest - tapping system. Types of tapping - Ladder, slaughter and puncture
tapping.
Tapping systems in India, yield stimulants and rain guarding
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 2

Explain the composition, preservation and concentration of NR latex

This module is presented from the official syllabus scope for Natural Rubber Production. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 2 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the composition, preservation and concentration of NR latex
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

2 Understanding Natural rubber latex Describe the various short term and long term

2 Understanding Natural rubber latex Describe the various short term and long term is an official topic in Module 2 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Natural rubber latex Describe the various short term and long term using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding natural rubber latex describe the various short term and long term should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Natural rubber latex Describe the various short term and long term means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 2 Understanding Natural rubber latex Describe the various short term and long term □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

4 Understanding preservative systems used in NR latex Explain the major concentration methods of 4

4 Understanding preservative systems used in NR latex Explain the major concentration methods of 4 is an official topic in Module 2 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 4 Understanding preservative systems used in NR latex Explain the major concentration methods of 4 using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 4 understanding preservative systems used in nr latex explain the major concentration methods of 4 should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 4 Understanding preservative systems used in NR latex Explain the major concentration methods of 4 means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ 4 Understanding preservative systems used in NR latex Explain the major concentration methods of 4 □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Understanding NR latex. Describe the production of value added

Understanding NR latex. Describe the production of value added is an official topic in Module 2 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify Understanding NR latex. Describe the production of value added using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, understanding nr latex. describe the production of value added should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What Understanding NR latex. Describe the production of value added means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 2-□□ Understanding NR latex. Describe the production of value added □□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

products from latex like double centrifuged 2 Understanding latex and skim rubber Definition of NR latex - composition and colloid nature - role of micro organisms in pre coagulation - short term and long term preservation systems - Anti coagulants, Concentration of NR latex - creaming, centrifuging, principle process machinery and plant lay out. BIS specification of centrifuged latex. Double centrifuged latex, skim rubber.

products from latex like double centrifuged 2 Understanding latex and skim rubber Definition of NR latex - composition and colloid nature - role of micro organisms in pre coagulation - short term and long term preservation systems - Anti coagulants, Concentration of NR latex - creaming, centrifuging, principle process machinery and plant lay out. BIS specification of centrifuged latex. Double centrifuged latex, skim rubber. is an official topic in Module 2 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify products from latex like double centrifuged 2 Understanding latex and skim rubber Definition of NR latex - composition and colloid nature - role of micro organisms in pre coagulation - short term and long term preservation systems - Anti coagulants, Concentration of NR latex - creaming, centrifuging, principle process machinery and plant lay out. BIS specification of centrifuged latex. Double centrifuged latex, skim rubber. using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, products from latex like double centrifuged 2 understanding latex and skim rubber definition of nr latex - composition and colloid nature - role of micro organisms in pre coagulation - short term and long term preservation systems - anti coagulants, concentration of nr latex - creaming, centrifuging, principle process machinery and plant lay out. bis specification of centrifuged latex. double centrifuged latex, skim rubber. should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What products from latex like double centrifuged 2 Understanding latex and skim rubber Definition of NR latex - composition and colloid nature - role of micro organisms in pre coagulation - short term and long term preservation systems - Anti coagulants, Concentration of NR latex - creaming, centrifuging, principle process machinery and plant lay out. BIS specification of centrifuged latex. Double centrifuged latex, skim rubber. means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 2-□□ products from latex like double centrifuged 2 Understanding latex and skim rubber Definition of NR latex - composition and colloid nature - role of micro organisms in pre coagulation - short term and long term preservation systems - Anti coagulants, Concentration of NR latex - creaming, centrifuging, principle process machinery and plant lay out. BIS specification of centrifuged latex. Double centrifuged latex, skim rubber. □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the composition, preservation and concentration of NR latex
            Explain the composition and colloid nature of
  M2.01                                                         2
Understanding
            Natural rubber latex
            Describe the various short term and long term
  M2.02                                                         4
Understanding
            preservative systems used in NR latex
            Explain the major concentration methods of          4
  M2.03
Understanding
            NR latex.
            Describe the production of value added
  M2.04     products from latex like double centrifuged         2
Understanding
            latex and skim rubber
            Series Test - I                                     1
Contents:
Definition of NR latex - composition and colloid nature - role of micro organisms
in pre
coagulation - short term and long term preservation systems - Anti coagulants,
Concentration of NR latex - creaming, centrifuging, principle process machinery and
plant
lay out. BIS specification of centrifuged latex. Double centrifuged latex, skim
rubber.
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 3

Explain the processing of NR latex in to dry marketable forms

This module is presented from the official syllabus scope for Natural Rubber Production. Use the topic cards for learning and the source transcript for exact coverage.

As specified
official hours

CO-linked
see outcomes

Learning goals

- Explain the official Module 3 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: Explain the processing of NR latex in to dry marketable forms
- Official duration: As specified
- Official topic entries captured: 4
- The full source excerpt is preserved below for coverage checking.

1 Remembering rubber Describe the machinery and process of

1 Remembering rubber Describe the machinery and process of is an official topic in Module 3 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Remembering rubber Describe the machinery and process of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 remembering rubber describe the machinery and process of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Remembering rubber Describe the machinery and process of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 3-□□ 1 Remembering rubber Describe the machinery and process of □□□□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□□.

3 Understanding manufacture of sheet rubber. Describe the machinery and process of

3 Understanding manufacture of sheet rubber. Describe the machinery and process of is an official topic in Module 3 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding manufacture of sheet rubber. Describe the machinery and process of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding manufacture of sheet rubber. describe the machinery and process of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding manufacture of sheet rubber. Describe the machinery and process of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding manufacture of sheet rubber. Describe the machinery and process of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

3 Understanding manufacture of crepe rubber. Describe the machinery and process of

3 Understanding manufacture of crepe rubber. Describe the machinery and process of is an official topic in Module 3 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding manufacture of crepe rubber. Describe the machinery and process of using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding manufacture of crepe rubber. describe the machinery and process of should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 3 Understanding manufacture of crepe rubber. Describe the machinery and process of means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ 3 Understanding manufacture of crepe rubber. Describe the machinery and process of □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

manufacture of technically specified block 3 Understanding rubber. Sheet rubber - Ribbed smoked sheets, Air dried sheets Processing, machinery, Grading - Green book. Crepe rubber - processing, machinery, different grades. ISNR-Processing, and machinery - grades - BIS specification. Describe the production, properties and applications of specialty rubbers

manufacture of technically specified block 3 Understanding rubber. Sheet rubber - Ribbed smoked sheets, Air dried sheets Processing, machinery, Grading - Green book. Crepe rubber - processing, machinery, different grades. ISNR-Processing, and machinery - grades - BIS specification. Describe the production, properties and applications of specialty rubbers is an official topic in Module 3 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify manufacture of technically specified block 3 Understanding rubber. Sheet rubber - Ribbed smoked sheets, Air dried sheets Processing, machinery, Grading - Green book. Crepe rubber - processing, machinery, different grades. ISNR-Processing, and machinery - grades - BIS specification. Describe the production, properties and applications of specialty rubbers using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, manufacture of technically specified block 3 understanding rubber. sheet rubber - ribbed smoked sheets, air dried sheets processing, machinery, grading - green book. crepe rubber - processing, machinery, different grades. isnr-processing, and machinery - grades - bis specification. describe the production, properties and applications of specialty rubbers should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What manufacture of technically specified block 3 Understanding rubber. Sheet rubber - Ribbed smoked sheets, Air dried sheets Processing, machinery, Grading - Green book. Crepe rubber - processing, machinery, different grades. ISNR-Processing, and machinery - grades - BIS specification. Describe the production, properties and applications of specialty rubbers means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 3-□□ manufacture of technically specified block 3 Understanding rubber. Sheet rubber - Ribbed smoked sheets, Air dried sheets Processing, machinery, Grading - Green book. Crepe rubber - processing, machinery, different grades. ISNR-Processing, and machinery - grades - BIS specification. Describe the production, properties and applications of specialty rubbers □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
Explain the processing of NR latex in to dry marketable forms
            List the different marketable forms of dry natural
  M3.01                                                           1
Remembering
            rubber
            Describe the machinery and process of
  M3.02                                                           3
Understanding
            manufacture of sheet rubber.
            Describe the machinery and process of
  M3.03                                                           3
Understanding
            manufacture of crepe rubber.
            Describe the machinery and process of
  M3.04     manufacture of technically specified block            3
Understanding
            rubber.

Contents:
Sheet rubber - Ribbed smoked sheets, Air dried sheets Processing, machinery,
Grading -
Green book. Crepe rubber - processing, machinery, different grades. ISNR-
Processing, and
machinery - grades - BIS specification.

            Describe the production, properties and applications of specialty
rubbers
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

MODULE 4

and reclaimed rubber

This module is presented from the official syllabus scope for Natural Rubber Production. Use the topic cards for learning and the source transcript for exact coverage.

As specified	**CO-linked**
official hours	see outcomes

Learning goals

- Explain the official Module 4 scope and its relationship to the course outcomes.
- Recognise the terminology, sequence, components and evidence expected in examination answers.
- Connect at least one official topic to a practical, industrial, laboratory or design context.

Key facts

- Official module title: and reclaimed rubber
- Official duration: As specified
- Official topic entries captured: 5
- The full source excerpt is preserved below for coverage checking.

1 Remembering Rubber. Explain the production of superior processing

1 Remembering Rubber. Explain the production of superior processing is an official topic in Module 4 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 1 Remembering Rubber. Explain the production of superior processing using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 1 remembering rubber. explain the production of superior processing should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 1 Remembering Rubber. Explain the production of superior processing means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ 1 Remembering Rubber. Explain the production of superior processing □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding rubbers Explain processing of Natural rubber latex into

2 Understanding rubbers Explain processing of Natural rubber latex into is an official topic in Module 4 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding rubbers Explain processing of Natural rubber latex into using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding rubbers explain processing of natural rubber latex into should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding rubbers Explain processing of Natural rubber latex into means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related

but different term.

Malayalam support: Module 4-□□ 2 Understanding rubbers Explain processing of Natural rubber latex into □□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

2 Understanding Constant viscosity and low viscosity rubbers. Describe the production of OENR and Tyre

2 Understanding Constant viscosity and low viscosity rubbers. Describe the production of OENR and Tyre is an official topic in Module 4 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding Constant viscosity and low viscosity rubbers. Describe the production of OENR and Tyre using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding constant viscosity and low viscosity rubbers. describe the production of oenr and tyre should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding Constant viscosity and low viscosity rubbers. Describe the production of OENR and Tyre means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

> **Exam tip:** Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 2 Understanding Constant viscosity and low viscosity rubbers. Describe the production of OENR and Tyre □□□□□□□□□□□□ □□□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□□ □□□□□□□ □□□□□□□□, steps, application □□□□□□□ □□□□□□□□□□ □□□□□□□. Technical terms English-□ □□□□□□ □□□□□□□□□□□□.

2 Understanding rubber. Describe digestor and reclamator process for the production of reclaimed rubber. Properties

2 Understanding rubber. Describe digestor and reclamator process for the production of reclaimed rubber. Properties is an official topic in Module 4 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 2 Understanding rubber. Describe digestor and reclamator process for the production of reclaimed rubber. Properties using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 2 understanding rubber. describe digestor and reclamator process for the production of reclaimed rubber. properties should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare
Definition/identity	What 2 Understanding rubber. Describe digestor and reclamator process for the production of reclaimed rubber. Properties means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 2 Understanding rubber. Describe digestor and reclamator process for the production of reclaimed rubber. Properties □□□□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

3 Understanding and application of hawai crumb rubber (buffed rubber Specialty rubbers - production - advantages and disadvantages - superior processing rubbers - PA 50 and PA 80, CV and LV rubbers, OENR, Tyre rubber. Reclaim rubber - List various production process, Reclamator process, Digester process - Advantages of Reclaim rubber Properties and application of hawai crumb rubber Text / Reference: T/R Book Title/Author R1 Hand book of natural rubber production -Rubber board High polymer lattices vol 1 -D C Blackley (Publisher - Springer Netherlands, 1997 R2 Edition) High Polymer Lattices, vol-I, vol-II, Vol-III - D.C.Blackley (Publisher - R3 SpringerNetherlands,1997) R4 Latex in Industry - R.J.Noble (Publisher - Rubber Age, Palmerton, New York). R5 Training Manual - Rubber Training Institute, RRII, Kottayam Rubbery Materials and their compounds - J.A Brydson(Publisher -Elsevier Applied R6 Science, London, 1988) Online Resources: Sl.No Website Link 1 http://www.rubberscience.in. 2 https://www.researchgate.net 3 https://www.scribd.com 4 http://dl.lib.mrt.ac.lk \5 http://www.indiannaturalrubber.com

3 Understanding and application of hawai crumb rubber (buffed rubber Specialty rubbers - production - advantages and disadvantages - superior processing rubbers - PA 50 and PA 80, CV and LV rubbers, OENR, Tyre rubber. Reclaim rubber - List various production process, Reclamator process, Digester process - Advantages of Reclaim rubber Properties and application of hawai crumb rubber Text / Reference: T/R Book Title/Author R1 Hand book of natural rubber production -Rubber board High polymer lattices vol 1 -D C Blackley (Publisher - Springer Netherlands, 1997 R2 Edition) High Polymer Lattices, vol-I, vol-II, Vol-III - D.C.Blackley (Publisher - R3 SpringerNetherlands,1997) R4 Latex in Industry - R.J.Noble (Publisher - Rubber Age, Palmerton, New York). R5 Training Manual - Rubber Training Institute, RRII, Kottayam Rubbery Materials and their compounds - J.A Brydson(Publisher -Elsevier Applied R6 Science, London, 1988) Online Resources: Sl.No Website Link 1 http://www.rubberscience.in. 2 https://www.researchgate.net 3 https://www.scribd.com 4 http://dl.lib.mrt.ac.lk \5 http://www.indiannaturalrubber.com is an official topic in Module 4 of Natural Rubber Production. Study the terminology first, then connect the stated purpose, sequence, components or application to the wider course outcome.

Concept and explanation

To learn this topic effectively, identify what the system, process, material, method or concept is; state its governing idea; describe the important parts or stages; and explain what changes when operating conditions change. The official source wording is retained in the module transcript so that examination answers remain aligned with the syllabus.

Study points

- Define or identify 3 Understanding and application of hawai crumb rubber (buffed rubber Specialty rubbers - production - advantages and disadvantages - superior processing rubbers - PA 50 and PA 80, CV and LV rubbers, OENR, Tyre rubber. Reclaim rubber - List various production process, Reclamator process, Digester process - Advantages of Reclaim rubber Properties and application of hawai crumb rubber Text / Reference: T/R Book Title/Author R1 Hand book of natural rubber production -Rubber board High polymer lattices vol 1 -D C Blackley (Publisher - Springer Netherlands, 1997 R2 Edition) High Polymer Lattices, vol-I, vol-II, Vol-III - D.C.Blackley (Publisher - R3 SpringerNetherlands,1997) R4 Latex in Industry - R.J.Noble (Publisher - Rubber Age, Palmerton, New York). R5 Training Manual - Rubber Training Institute, RRII, Kottayam Rubbery Materials and their compounds - J.A Brydson(Publisher -Elsevier Applied R6 Science, London, 1988) Online Resources: Sl.No Website Link 1 http://www.rubberscience.in. 2 https://www.researchgate.net 3 https://www.scribd.com 4 http://dl.lib.mrt.ac.lk \5 http://www.indiannaturalrubber.com using standard technical terminology.
- Arrange the important components, steps or characteristics in a logical sequence.
- Relate the topic to one laboratory, industrial, professional or design situation where appropriate.
- Use a labelled sketch, comparison table, formula or procedure when the question requires it.

Engineering or professional relevance

In Polytechnic practice, 3 understanding and application of hawai crumb rubber (buffed rubber specialty rubbers - production - advantages and disadvantages - superior processing rubbers - pa 50 and pa 80, cv and lv rubbers, oenr, tyre rubber. reclaim rubber - list various production process, reclamator process, digester process - advantages of reclaim rubber properties and application of hawai crumb rubber text / reference: t/r book title/author r1 hand book of natural rubber production -rubber board high polymer lattices vol 1 -d c blackley (publisher - springer netherlands, 1997 r2 edition) high polymer lattices, vol-i, vol-ii, vol-iii - d.c.blackley (publisher - r3 springernetherlands,1997) r4 latex in industry - r.j.noble (publisher - rubber age, palmerton, new york). r5 training manual - rubber training institute, rrii, kottayam rubbery materials and their compounds - j.a brydson(publisher -elsevier applied r6 science, london, 1988) online resources: sl.no website link 1 http://www.rubberscience.in. 2 https://www.researchgate.net 3 https://www.scribd.com 4 http://dl.lib.mrt.ac.lk \5 http://www.indiannaturalrubber.com should be discussed with attention to safe operation, correct terminology, observable evidence and the relationship between input, process and result.

Topic study guide

Aspect	What to prepare

Aspect	What to prepare
Definition/identity	What 3 Understanding and application of hawai crumb rubber (buffed rubber Specialty rubbers - production - advantages and disadvantages - superior processing rubbers - PA 50 and PA 80, CV and LV rubbers, OENR, Tyre rubber. Reclaim rubber - List various production process, Reclamator process, Digester process - Advantages of Reclaim rubber Properties and application of hawai crumb rubber Text / Reference: T/R Book Title/Author R1 Hand book of natural rubber production -Rubber board High polymer lattices vol 1 -D C Blackley (Publisher - Springer Netherlands, 1997 R2 Edition) High Polymer Lattices, vol-I, vol-II, Vol-III - D.C.Blackley (Publisher - R3 SpringerNetherlands,1997) R4 Latex in Industry - R.J.Noble (Publisher - Rubber Age, Palmerton, New York). R5 Training Manual - Rubber Training Institute, RRII, Kottayam Rubbery Materials and their compounds - J.A Brydson(Publisher -Elsevier Applied R6 Science, London, 1988) Online Resources: Sl.No Website Link 1 http://www.rubberscience.in. 2 https://www.researchgate.net 3 https://www.scribd.com 4 http://dl.lib.mrt.ac.lk \5 http://www.indiannaturalrubber.com means in the official course context
Study lens	Principle → components or stages → result → application
Exam evidence	Definition, labelled explanation, comparison, procedure or example as appropriate

Exam tip: Begin with the exact term, then give an organised explanation with a diagram, table, procedure or example whenever the topic naturally supports it.

Common mistakes: Writing a memorised phrase without explaining the principle; omitting units, labels, conditions or sequence; or mixing this topic with a related but different term.

Malayalam support: Module 4-□□ 3 Understanding and application of hawai crumb rubber (buffed rubber Specialty rubbers - production - advantages and disadvantages - superior processing rubbers - PA 50 and PA 80, CV and LV rubbers, OENR, Tyre rubber. Reclaim rubber - List various production process, Reclamator process, Digester process - Advantages of Reclaim rubber Properties and application of hawai crumb rubber Text / Reference: T/R Book Title/Author R1 Hand book of natural rubber production -Rubber board High polymer lattices vol 1 -D C Blackley (Publisher - Springer Netherlands, 1997 R2 Edition) High Polymer Lattices, vol-I, vol-II, Vol-III - D.C.Blackley (Publisher - R3 SpringerNetherlands,1997) R4 Latex in Industry - R.J.Noble (Publisher - Rubber Age, Palmerton, New York). R5 Training Manual - Rubber Training Institute, RRII, Kottayam Rubbery Materials and their compounds - J.A Brydson(Publisher -Elsevier Applied R6 Science, London, 1988) Online Resources: Sl.No Website Link 1 http://www.rubberscience.in. 2 https://www.researchgate.net 3 https://www.scribd.com 4 http://dl.lib.mrt.ac.lk \5 http://www.indiannaturalrubber.com □□□□□□□□□□□ □□□□□ definition/meaning □□□□□□□□□□□□□□. □□□□□□ □□□□□□ □□□□□□□, steps, application □□□□□□ □□□□□□□□□ □□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Official syllabus detail and terminology

The following excerpt preserves official syllabus terminology for this module. Educational explanations below are separate elaboration.

```
and reclaimed rubber
            List the different specialty forms of Natural
  M4.01                                                         1
Remembering
            Rubber.
            Explain the production of superior processing
  M4.02                                                         2
Understanding
            rubbers
            Explain processing of Natural rubber latex into
  M4.03                                                         2
Understanding
            Constant viscosity and low viscosity rubbers.
            Describe the production of OENR and Tyre
  M4.04                                                         2
Understanding
            rubber.
            Describe digestor and reclamator process for
            the production of reclaimed rubber. Properties
  M4.05                                                         3
Understanding
            and application of hawai crumb rubber
            (buffed rubber
            Series Test - II                                    1

Contents:
Specialty rubbers - production - advantages and disadvantages - superior
```

Module study routine: Read the source wording, make a one-page concept map, practise one short answer and one structured answer, then review the Malayalam support notes.

RAPID REFERENCE

Concept and formula bank

Answer structure

```
Definition → Principle → Components/steps → Application → Conclusion
```

Use this sequence for descriptive questions when the topic does not require a numerical formula.

Practical record

```
Aim → Apparatus → Procedure → Observation → Result → Safety
```

Use this sequence for laboratory, workshop and field activities.

RAPID REVISION

Diagram and source-transcript library

Module 1 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 2 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 3 source and topic cards

Jump to official terminology, topic explanations and study guidance.

Module 4 source and topic cards

Jump to official terminology, topic explanations and study guidance.

OFFICIAL SELF-LEARNING

Activities from the syllabus

Use the extracted official activity wording as the record of required self-learning work.

The official PDF does not expose a separate activities heading in the extracted text; consult the source PDF pages for the exact activity list.

EXAMINATION PREPARATION

Assessment methodology

The following source extract preserves the assessment information detected in the official PDF. Verify mark conversions and institutional updates against the current source before an examination.

```
The official assessment structure is reproduced from the supplied PDF.
```

Practice-paper notice: The model paper below is generated for revision and is not an official examination paper.

ACTIVE RECALL

Module-wise questions and answers

These are practice questions based on official topic coverage, not official examination questions.

Module 1

Define or explain Explain the importance and uniqueness of NR 2 Remembering Describe various propagation methods and. (3 marks)

Answer: Begin with the standard definition or identity of *Explain the importance and uniqueness of NR 2 Remembering Describe various propagation methods and*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain classification of clones in terms of rubber Understanding 3 plants Explain tapping and various processes associated. (3 marks)

Answer: Begin with the standard definition or identity of *classification of clones in terms of rubber Understanding 3 plants Explain tapping and various processes associated*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding with extraction of latex from rubber trees. Explain method of application and importance. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding with extraction of latex from rubber trees. Explain method of application and importance*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding of rain guarding and yield stimulants. Natural rubber - uniqueness, structure and properties, major sources statistics of production and consumption - India and World. Rubber yielding plants - propagation methods - brown and green budding - different clones, tapping - standard of tappability - tapping task - tapping rest - tapping system. Types of tapping - Ladder, slaughter and puncture tapping. Tapping systems in India, yield stimulants and rain guarding. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding of rain guarding and yield stimulants. Natural rubber - uniqueness, structure and properties,* *major sources statistics of production and consumption - India and World. Rubber yielding plants - propagation methods - brown and green budding - different clones, tapping - standard of tappability - tapping task - tapping rest - tapping system. Types of tapping - Ladder, slaughter and puncture tapping. Tapping systems in India, yield stimulants and rain guarding*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 2

Define or explain 2 Understanding Natural rubber latex Describe the various short term and long term. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding Natural rubber latex Describe the various short term and long term*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 4 Understanding preservative systems used in NR latex Explain the major concentration methods of 4. (3 marks)

Answer: Begin with the standard definition or identity of *4 Understanding preservative systems used in NR latex Explain the major concentration methods of 4*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain Understanding NR latex. Describe the production of value

added. (3 marks)

Answer: Begin with the standard definition or identity of *Understanding NR latex. Describe the production of value added*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain products from latex like double centrifuged 2 Understanding latex and skim rubber Definition of NR latex - composition and colloid nature - role of micro organisms in pre coagulation - short term and long term preservation systems - Anti coagulants, Concentration of NR latex - creaming, centrifuging, principle process machinery and plant lay out. BIS specification of centrifuged latex. Double centrifuged latex, skim rubber.. (3 marks)

Answer: Begin with the standard definition or identity of *products from latex like double centrifuged 2 Understanding latex and skim rubber Definition of NR latex - composition and colloid nature - role of micro organisms in pre coagulation - short term and long term preservation systems - Anti coagulants, Concentration of NR latex - creaming, centrifuging, principle process machinery and plant lay out. BIS specification of centrifuged latex. Double centrifuged latex, skim rubber.*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 3

Define or explain 1 Remembering rubber Describe the machinery and process of. (3 marks)

Answer: Begin with the standard definition or identity of *1 Remembering rubber Describe the machinery and process of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding manufacture of sheet rubber. Describe the machinery and process of. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding manufacture of sheet rubber. Describe the machinery and process of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 3 Understanding manufacture of crepe rubber. Describe the machinery and process of. (3 marks)

Answer: Begin with the standard definition or identity of *3 Understanding manufacture of crepe rubber. Describe the machinery and process of*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

Malayalam support: □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain manufacture of technically specified block 3 Understanding rubber. Sheet rubber - Ribbed smoked sheets, Air dried sheets Processing, machinery, Grading - Green book. Crepe rubber - processing, machinery, different grades. ISNR-Processing, and machinery - grades - BIS specification. Describe the production, properties and applications of specialty rubbers. (3 marks)

Answer: Begin with the standard definition or identity of *manufacture of technically specified block 3 Understanding rubber. Sheet rubber - Ribbed smoked sheets, Air dried sheets Processing, machinery, Grading - Green book. Crepe rubber - processing, machinery, different grades. ISNR-Processing, and machinery - grades - BIS specification. Describe the production, properties and applications of specialty rubbers*.

State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Module 4

Define or explain 1 Remembering Rubber. Explain the production of superior processing. (3 marks)

Answer: Begin with the standard definition or identity of *1 Remembering Rubber. Explain the production of superior processing*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding rubbers Explain processing of Natural rubber latex into. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding rubbers Explain processing of Natural rubber latex into*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding Constant viscosity and low viscosity rubbers. Describe the production of OENR and Tyre. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding Constant viscosity and low viscosity rubbers. Describe the production of OENR and Tyre*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

Define or explain 2 Understanding rubber. Describe digestor and reclamator process for the production of reclaimed rubber. Properties. (3 marks)

Answer: Begin with the standard definition or identity of *2 Understanding rubber. Describe digestor and reclamator process for the production of reclaimed rubber. Properties*. State the governing principle, list the key components or stages, and close with one relevant application or observation. Use the exact official terminology preserved in the module transcript.

> **Malayalam support:** □ □□□□□□□□□□□ □□□□□ definition, □□□□□□ principle/steps, □□□□□□ application □□□□ □□□□□□□□□ □□□□□□.

PRACTICE ONLY

Practice Model Paper — Not an Official Examination Paper

Attempt one short definition, one structured explanation and one long answer from every module. Use the official assessment pattern reproduced above when selecting time and marks.

Module 1

1-mark definition: State the meaning of **Explain the importance and uniqueness of NR 2 Remembering Describe various propagation methods and**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 2

1-mark definition: State the meaning of **2 Understanding Natural rubber latex Describe the various short term and long term**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 3

1-mark definition: State the meaning of **1 Remembering rubber Describe the machinery and process of**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

Module 4

1-mark definition: State the meaning of **1 Remembering Rubber. Explain the production of superior processing**.

3-mark explanation: Explain one principle, process or comparison from this module.

7-mark answer: Write a structured answer using a labelled diagram, table, procedure, example or application.

QUICK REVISION

Exam checklist

Before writing

- Read the command word: define, explain, compare, draw, calculate, analyse or design.
- Use the exact course terminology from the source transcript.

- Plan labels, units, sequence and marks before writing.

Before submitting

- Check every module has been revised.
- Check diagrams, tables, formula symbols and units.
- Separate official syllabus information from personal elaboration.

REFERENCE VOCABULARY

Glossary

Study glossary

Term	Meaning in this lesson
Course Outcome	A capability the student should demonstrate after completing the course.
Module	An official syllabus unit with defined topics and hours.
CIE	Continuous Internal Evaluation.
ESE	End Semester Examination.
Source transcript	A preserved extract of the official syllabus used for coverage checking.

SOURCE-SUPPORTED REFERENCES

References and web resources

References listed in the official PDF

References are included in the official source PDF.

Web resources listed in the official PDF

- Sl.No Website Link
- http://www.rubberscience.in. https://www.researchgate.net https://www.scribd.com http://dl.lib.mrt.ac.lk \5 http://www.indiannaturalrubber.com

IN-PAGE SEARCH

Search this lesson